

Lihsuan Chuang

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Vancouver, BC, Canada

RESEARCH INTERESTS

My research interest spans across computer architecture and security with a focus on microarchitectural side-channel attacks. My current work explores keystroke inference attacks on Arm-based hardware and high-performance microarchitectural weird machines.

EDUCATION

- **University of British Columbia** 09/2026 – 06/2030 (expected)
Doctor of Philosophy in Electrical and Computer Engineering — Advisor: Prof. Prashant Nair
Vancouver, Canada
- **University of North Carolina at Chapel Hill** 09/2024 – 06/2026
Master of Science in Computer Science — Advisor: Prof. Andrew Kwong
Chapel Hill, United States
- **University of British Columbia** 09/2020 – 06/2024
Bachelor of Applied Science in Computer Engineering with Distinction
Vancouver, Canada

PUBLICATIONS

- **KeyTAR: Practical Keystroke Timing Attacks and Input Reconstruction** 09/2024 – 09/2025
Mufan Qiu*, Lihsuan Chuang*, Dohhyun Kim, Huaizhi Qu, Tianlong Chen, Andrew Kwong *Accepted at IEEE S&P 2026*
 - Designed and demonstrated the **first end-to-end input-reconstruction attack** from inter-keystroke timing signals, effective in both native and browser environments
 - Built a dataset collection pipeline to replay typing samples and capture high-fidelity side-channel traces at scale
 - Implemented the first general Prime+Probe keystroke extraction attack without the need of a target

AWARDS & FELLOWSHIPS

- **IEEE S&P Student Travel Grant** 04/2026
IEEE Symposium on Security and Privacy 2026
San Francisco, United States
- **UBC Four-Year Doctoral Fellowship** 03/2026
Awarded upon PhD Admission — University of British Columbia
Vancouver, Canada
- **UBC Outstanding International Student Award** 03/2020
Awarded upon Bachelor Admission — University of British Columbia
Vancouver, Canada

INVITED TALKS & PRESENTATIONS

- **IEEE Symposium on Security and Privacy 2026 — KeyTAR** 05/2026
Conference Proceedings Presentation
San Francisco, United States
- **UNC COMP 590-184 Hardware Security — KeyTAR** 03/2026
Guest Lecture
Chapel Hill, United States
- **UNC COMP 560 Artificial Intelligence — KeyTAR** 02/2026
Guest Lecture
Chapel Hill, United States

RESEARCH & PROFESSIONAL EXPERIENCE

- **University of North Carolina at Chapel Hill — Dept. of Computer Science** 09/2024 – present
Graduate Research Assistant, Advisor: Prof. Andrew Kwong
Chapel Hill, United States
 - Led the full research cycle for *KeyTAR*: threat modeling, dataset collection, attack implementation, and evaluation against real-world targets
 - Investigated into the feasibility of keystroke extraction attacks on Arm-based hardware devices
- **IEEE Symposium on Security and Privacy 2026 [web]** 08/2025, 05/2026
External Reviewer and Presenter
Oakland, United States
 - Reviewed submissions in microarchitectural security and side-channel attacks
 - Provided detailed feedback on organization, scientific value, novelty, and technical correctness
 - Presented *KeyTAR* to an international audience of security researchers at one of the premier venues in computer security (IEEE S&P)
- **ACM Conference on Computer and Communications Security 2025 [web]** 10/2025
Attendee
Taipei, Taiwan

- Participated in workshops and presentations of published works
- Exchanged research ideas with security researchers from globally renowned institutions

- **UBC Student Housing and Community Services** ^[web]

09/2022 – 06/2024

Residence Advisor

Vancouver, Canada

- Enforced residence standards and expectations with assertive communication
- Organized events to foster residents involvement in the community
- Coordinated with peers and supervisors to complete assigned tasks

- **UBC Thunderbots** ^[web]

09/2021 – 06/2024

Software Engineer

Vancouver, Canada

- Supported integration of a physics-accurate game simulator
- Developed test suites validating the effectiveness of robot tactics
- Implemented a fault-detection and robot-substitution algorithm

- **iNet Internship**

06/2021 – 08/2021

Web Developer

Taipei, Taiwan

- Designed interactive web pages integrating popular third-party widgets
- Performed systematic testing to surface edge-case bugs from unexpected user behavior

TEACHING EXPERIENCE

- **University of North Carolina at Chapel Hill – COMP 590-184: Hardware Security**

01/2026 – 05/2026

Teaching Assistant, Advisor: Prof. Andrew Kwong

Chapel Hill, United States

- Responded to student technical inquiries on Campuswire
- Hosted weekly in-person office hours covering hardware security concepts
- Delivered two guest lectures on keystroke timing side-channel attacks, bridging course theory with cutting-edge published research

- **University of British Columbia – Department of Electrical and Computer Engineering**

09/2022 – 06/2024

Teaching Assistant (APSC 160, CPEN 211, CPEN 212)

Vancouver, Canada

- Led tutorial sessions reinforcing core concepts in introductory programming and digital systems design
- Graded coursework and exams to support instructor course management
- Collaborated with instructors to improve overall course delivery and assessment quality

SELECTED PROJECTS

- **KeyTAR: Practical Keystroke Timing Attacks and Input Reconstruction**

09/2024–06/2025

Tools: C, Javascript, Python, NodeJS

[\[code\]](#)

- Implemented an automated cache trace collection tool on native and the browser environment

- **HistoData: Automated Trading System and Visualization Tool**

09/2023–06/2024

Tools: C#, ReactJS, Python, NodeJS, SQL, Interactive Broker API

[\[code\]](#)

- Implemented an automated trading data visualization desktop tool
- Built middleware to bridge a ReactJS front end to the Interactive Broker API

- **SDMS: Smart Doorbell System**

01/2022–05/2022

Tools: Verilog, Lua, Flutter, AWS

[\[code\]](#)

- Integrated hardware, mobile application, and cloud server for a smart doorbell
- Established cross-team communication protocols to ensure cohesive system integration

- **RISC Computer**

01/2021–05/2021

Tools: DE1-SOC & Verilog

[\[code\]](#)

- Designed and implemented a RISC processor (ALU, registers, datapath, memory) capable of executing a subset of ARMv7 instructions using HDL

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python, R, Matlab, Verilog, JavaScript, HTML, CSS
- **Security & Systems:** Side-channel analysis, microarchitectural attacks, hardware security, embedded systems, HDL design
- **Languages:** Mandarin Chinese (Native), English (Native), Japanese (JLPT N2, 12/2024), French (Intermediate)